Technical Documentation

Universal Drone Adapter Board

CM4 & CM5 Nano-B Compatible Interface Board

Hardware Revision:	Version 4.0 - JST GH Integration
Target Platforms:	Waveshare CM4-NANO-B, Waveshare CM5-NANO-B
Application:	Drone Sensor Hub, Pixhawk Telemetry Interface, Power Routing
Organization:	PSYC Aerospace and Defence Industries Pvt. Ltd.
Document Type:	Hardware Technical Reference Manual

Designed & Manufactured in India
PSYC Aerospace and Defence Industries Pvt. Ltd.

Contents

1 System Overview

The Universal Drone Adapter Board is a compact interface PCB designed to convert the standard 40-pin GPIO header of the Waveshare CM4-NANO-B and CM5-NANO-B carrier boards into drone-ready peripheral connectors.

The board acts as a reliable translation and breakout layer between the Raspberry Pi Compute Module platform and external drone electronics such as Pixhawk flight controllers, GPS modules, optical flow sensors, compass modules, SPI sensors, and auxiliary payload controllers.

The design replaces direct jumper wiring with vibration-resistant JST GH 1.25 mm connectors and clearly labelled test pads. This makes the board suitable for UAV prototyping, embedded flight-computer integration, GPS-denied navigation systems, sensor fusion payloads, and companion-computer telemetry links.

1.1 Primary Functions

- Breaks out UART, I2C, SPI, 5 V, 3.3 V, and GND lines from the CM4/CM5 Nano-B 40-pin header.
- Provides JST GH connectors for drone-grade wiring harnesses.
- Supports Pixhawk telemetry connection through dedicated UART ports.
- Supports I2C sensor buses for GPS compass, optical flow, range sensors, and payload electronics.
- Supports SPI peripherals through a dedicated SPI breakout port.
- Provides onboard safe-shutdown input through GPIO21.
- Provides onboard status LED output through GPIO6.
- Provides configurable 5 V and 3.3 V rail access through jumpers and test pads.

2 Board Architecture

2.1 Functional Block Diagram

External Drone Hardware
Pixhawk FC, GPS, Compass, Optical Flow, Payload MCU, SPI Sensors

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Universal Drone Adapter Board V4.0
JST GH UART Ports, I2C Ports, SPI Port, 40-pin Header, Power Jumpers, GPIO21 Shutdown, GPIO6 LED

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Waveshare CM4-NANO-B / CM5-NANO-B Carrier Board
Raspberry Pi Compute Module 4 or Compute Module 5 Companion Computer

2.2 Mechanical and Connector Layout

The PCB uses a rectangular mounting pattern with four mounting holes, allowing the board to be fixed above or beside the Compute Module carrier inside a UAV avionics bay.

The top-mounted JST GH connectors are positioned for easy harness routing. The center of the board contains the 40-pin GPIO header interface. The bottom side of the board provides additional labelled test pads and redundant signal access for debugging.

2.3 Target Use Cases

- Pixhawk companion-computer telemetry interface.
- Vision positioning system interface board.
- GPS replacement or GPS-denied navigation module.
- Drone sensor hub for I2C and SPI sensors.
- Payload controller interface for winch, camera gimbal, or mission module.
- UAV testbed wiring simplification board.

3 Electrical Overview

3.1 Logic Levels

Table 1: Electrical Logic Levels

Signal Type	Voltage Level	Notes
GPIO / UART / I2C / SPI Logic	3.3 V	Raspberry Pi GPIO pins are not 5 V tolerant.
Peripheral Power Output	5 V and 3.3 V	Available on selected JST and test-pad headers.
Ground Reference	Common GND	Must be shared between CM board, Pixhawk, sensors, and payload electronics.

Important: The UART, I2C, SPI, and GPIO signals must remain at 3.3 V logic level. Do not connect 5 V logic devices directly to the Raspberry Pi GPIO pins without level shifting.

3.2 Power Distribution

The board provides both 5 V and 3.3 V distribution points. The 5 V rail may be used to power external peripherals or to back-feed the Compute Module carrier, depending on jumper configuration and system wiring.

Table 2: Power Rails

Rail	Nominal Voltage	Purpose
+5V	5.0 V to 5.2 V	Peripheral power, Pixhawk-style connector supply, optional carrier input rail.
+3V3	3.3 V	Logic rail for I2C/SPI/UART peripherals requiring 3.3 V supply.
GND	0 V	Common return path for all ports and external modules.

3.3 Power Safety Notes

- Use a regulated 5 V BEC with sufficient current capacity when powering the system from a drone battery.
- Recommended external supply range for the 5 V rail is 5.0 V to 5.2 V.
- Do not inject 5 V from multiple sources unless reverse-current protection is installed and verified.
- Verify common ground between Pixhawk, Compute Module carrier, sensors, and this adapter board.
- Add a Schottky diode or ideal-diode protection if the board is used to back-power the Nano-B carrier.

4 Main 40-Pin Header Interface

The center header interfaces with the Raspberry Pi style 40-pin expansion connector available on the Waveshare CM4-NANO-B and CM5-NANO-B carrier boards.

Table 3: 40-Pin Header Signal Mapping Used by Adapter Board

Function	BCM GPIO	Physical Pin	Board Usage
I2C SDA	GPIO2	Pin 3	I2C bus data line for I2C sensor ports.
I2C SCL	GPIO3	Pin 5	I2C bus clock line for I2C sensor ports.
UART TXD0	GPIO14	Pin 8	UART Port 1 transmit line.
UART RXD0	GPIO15	Pin 10	UART Port 1 receive line.
UART Aux TX	GPIO4	Pin 7	UART Port 2 transmit line, software overlay dependent.
UART Aux RX	GPIO5	Pin 29	UART Port 2 receive line, software overlay dependent.
UART Aux TX	GPIO12	Pin 32	UART Port 3 transmit line, software overlay dependent.

Function	BCM GPIO	Physical Pin	Board Usage
UART Aux RX	GPIO13	Pin 33	UART Port 3 receive line, software overlay dependent.
SPI MOSI	GPIO10	Pin 19	SPI master output line.
SPI MISO	GPIO9	Pin 21	SPI master input line.
SPI SCLK	GPIO11	Pin 23	SPI clock line.
SPI CE0	GPIO8	Pin 24	SPI chip-enable 0.
SPI CE1	GPIO7	Pin 26	SPI chip-enable 1.
Status LED	GPIO6	Pin 31	Onboard status / heartbeat LED.
Shutdown Switch	GPIO21	Pin 40	Safe shutdown input switch.
+5V	–	Pins 2 / 4	5 V rail access.
+3V3	–	Pins 1 / 17	3.3 V rail access.
GND	–	Multiple pins	Common ground reference.

5 Connector Pinout

5.1 UART Connectors

The board provides three UART connector groups for telemetry, GPS, and auxiliary payload communication.

Table 4: UART Port Summary

Port	Signals	GPIO Lines	Primary Use
UART Port 1	5V, TX, RX, GND	GPIO14 / GPIO15	Primary Pixhawk telemetry / MAVLink serial link.
UART Port 2	5V, TX, RX, GND	GPIO4 / GPIO5	Auxiliary flight-controller link, payload MCU, debug serial.
UART Port 3	5V, TX, RX, GND	GPIO12 / GPIO13	GPS data link, external controller, payload serial.

5.1.1 UART Wiring Direction

When connecting the adapter board to a Pixhawk or external serial device:

Adapter Board TX	Connects to external device RX.
Adapter Board RX	Connects to external device TX.
Adapter Board GND	Connects to external device GND.
Adapter Board 5V	Use only if the external device is intended to receive 5 V from this port.

Warning: Pixhawk TELEM ports typically expose 5 V, TX, RX, CTS, RTS, and GND on a 6-pin JST GH connector. This board uses simplified UART wiring. Confirm the harness pin order before plugging into a flight controller.

5.2 I2C Connectors

The board provides three I2C breakout connector groups labelled I2C1, I2C2, and I2C3. These are intended for GPS compass, optical flow sensors, range sensors, and low-speed sensor modules.

Table 5: I2C Port Summary

Port	Signals	Primary Use
I2C Port 1	5V, 3V3, SDA, SCL, GND	Main I2C sensor bus.
I2C Port 2	5V, 3V3, SDA, SCL, GND	Secondary I2C sensor breakout.
I2C Port 3	5V, 3V3, SDA, SCL, GND	External compass / optical flow / auxiliary sensor port.

5.2.1 I2C Bus Notes

- SDA is routed from GPIO2.
- SCL is routed from GPIO3.

- All I2C devices on the same bus must have unique addresses.
- Keep I2C cable length short in high-vibration UAV environments.
- For long cables, reduce bus speed to 100 kHz and use shielded/twisted wiring where possible.
- Use 3.3 V logic I2C devices unless level shifters are installed.

5.3 SPI Connector

The SPI breakout is provided for high-speed peripherals such as ADCs, IMUs, flash memory, radio modules, and custom payload electronics.

Table 6: SPI Port Pinout

Signal	**BCM GPIO**	**Description**
MOSI	GPIO10	Master Out, Slave In.
MISO	GPIO9	Master In, Slave Out.
SCLK	GPIO11	SPI clock.
CE0	GPIO8	Chip enable 0.
CE1	GPIO7	Chip enable 1.
3V3	–	3.3 V peripheral supply.
5V	–	5 V peripheral supply.
GND	–	Ground reference.

6 Onboard Controls

6.1 Safe Shutdown Switch

The board includes a tactile switch connected to GPIO21. This input can be configured using the Raspberry Pi `gpio-shutdown` overlay to safely shut down Linux before removing power.

Table 7: Shutdown Switch Interface

Signal	GPIO21
Physical Pin	Pin 40
Function	Safe operating-system shutdown input.
Recommended Logic	Active-low with internal pull-up.

6.2 Status LED

The board includes an onboard LED connected to GPIO6. It may be configured as a heartbeat LED, system-alive indicator, disk activity monitor, or custom application status output.

Table 8: Status LED Interface

Signal	GPIO6
Physical Pin	Pin 31
Function	Status LED / heartbeat indicator.
Recommended Series Resistor	330 Ω to 1 kΩ, depending on LED brightness.

7 Software Configuration

Although the CM4 and CM5 Nano-B carrier boards use the same 40-pin header form factor, the internal peripheral routing differs between Raspberry Pi Compute Module 4 and Compute Module 5 platforms.

Therefore, the correct `config.txt` overlay must be selected based on the module installed.

7.1 CM4 Nano-B Configuration

For Raspberry Pi Compute Module 4, add the following lines to:

`/boot/config.txt`

```
# ==========================================
# Universal Drone Adapter Board - CM4 Config
# Hardware Revision: V4.0
# ==========================================

# Enable primary UART for UART Port 1
dtparam=uart0=on

# Enable UART3 on GPIO4/GPIO5 for UART Port 2
dtoverlay=uart3,ctsrts=off

# Enable UART5 on GPIO12/GPIO13 for UART Port 3
dtoverlay=uart5,ctsrts=off

# Enable I2C1 for I2C sensor ports
dtparam=i2c_arm=on

# Enable SPI0 for SPI breakout port
dtparam=spi=on

# Enable safe shutdown button on GPIO21
dtoverlay=gpio-shutdown,gpio_pin=21,active_low=1,gpio_pull=up

# Optional: map activity/status LED to GPIO6
dtparam=act_led_gpio=6
```

7.2 CM5 Nano-B Configuration

For Raspberry Pi Compute Module 5, add the following lines to:

`/boot/firmware/config.txt`

```
# ==========================================
# Universal Drone Adapter Board - CM5 Config
# Hardware Revision: V4.0
# ==========================================

# Enable primary UART for UART Port 1
dtparam=uart0=on

# Enable UART2 on GPIO4/GPIO5 for UART Port 2
# CM5/RP1 platform uses Pi 5 UART overlays
dtoverlay=uart2-pi5

# Enable UART4 on GPIO12/GPIO13 for UART Port 3
# CM5/RP1 platform uses Pi 5 UART overlays
dtoverlay=uart4-pi5

# Enable I2C1 for I2C sensor ports
dtparam=i2c_arm=on

# Enable SPI0 for SPI breakout port
dtparam=spi=on

# Enable safe shutdown button on GPIO21
dtoverlay=gpio-shutdown,gpio_pin=21,active_low=1,gpio_pull=up

# Optional: map activity/status LED to GPIO6
dtparam=act_led_gpio=6
```

7.3 Disable Linux Serial Console

For MAVLink or external telemetry use, the serial console should be disabled on the selected UART. Run:

```
sudo raspi-config
```

Then go to:

`Interface Options -> Serial Port`

Select:

- Login shell over serial: **No**
- Serial port hardware enabled: **Yes**

Reboot after changes:

```
sudo reboot
```

8 Linux Device Verification

8.1 Check UART Devices

After rebooting, check available serial devices:

```
ls -l /dev/ttyAMA*
ls -l /dev/serial*
```

Expected devices may vary depending on Raspberry Pi OS version and overlay support.

Table 9: Expected UART Devices

Platform	Expected Devices	Notes
CM4	`/dev/ttyAMA0`, auxiliary `ttyAMA` devices	Exact numbering depends on overlays and Bluetooth configuration.
CM5	`/dev/ttyAMA0`, `/dev/ttyAMA2`, `/dev/ttyAMA4` or similar	RP1 UART mapping may differ by OS release.

8.2 Check I2C Bus

Install tools:

```
sudo apt update
sudo apt install -y i2c-tools
```

Scan I2C bus:

```
sudo i2cdetect -y 1
```

8.3 Check SPI Interface

Check SPI devices:

```
ls -l /dev/spidev*
```

Expected output:

```
/dev/spidev0.0
/dev/spidev0.1
```

9 Pixhawk Integration

9.1 Recommended Pixhawk TELEM Connection

For Pixhawk telemetry, use UART Port 1 as the primary MAVLink connection.

Table 10: Pixhawk Telemetry Wiring

Adapter Board UART TX	Connect to Pixhawk TELEM RX.
Adapter Board UART RX	Connect to Pixhawk TELEM TX.
Adapter Board GND	Connect to Pixhawk TELEM GND.
Adapter Board 5V	Use only if powering requirement is confirmed. Avoid dual-power conflicts.

9.2 MAVLink Test

Install MAVProxy:

```
sudo apt install -y python3-pip
pip3 install MAVProxy
```

Test connection:

```
mavproxy.py --master=/dev/ttyAMA0 --baudrate 57600
```

Common baud rates:

- 57600
- 115200
- 921600
- 1500000

10 Manufacturing and Assembly Notes

10.1 Connector Recommendation

Table 11: Connector System

Connector Type	JST GH 1.25 mm locking connector
Use Case	UAV, Pixhawk, GPS, compass, optical flow, telemetry wiring
Advantage	Compact, vibration-resistant, keyed/locking harness interface

10.2 PCB Assembly Guidelines

- Verify JST connector orientation before production.
- Keep silkscreen pin labels visible after connector placement.
- Ensure 5 V and 3.3 V pins are clearly separated in layout.
- Avoid routing high-current power through thin signal traces.
- Add test pads for UART TX/RX, I2C SDA/SCL, SPI lines, 5 V, 3.3 V, and GND.
- Use mounting holes with adequate copper keepout to prevent shorts through metal standoffs.

11 Pre-Flight Verification Checklist

11.1 Visual Inspection

- Confirm there are no solder bridges between adjacent JST pins.
- Confirm the 40-pin header is aligned correctly.
- Confirm all mounting holes are mechanically clear.
- Confirm polarity markings for 5 V, 3.3 V, and GND.

11.2 Power Check

- Measure 5 V rail before connecting the Compute Module carrier.
- Confirm voltage is between 5.0 V and 5.2 V.
- Measure 3.3 V rail.
- Check continuity between all GND points.
- Confirm there is no short between 5 V and GND.
- Confirm there is no short between 3.3 V and GND.

11.3 Signal Check

- Verify UART TX/RX crossover before connecting to Pixhawk.
- Verify I2C SDA/SCL continuity to GPIO2/GPIO3.
- Verify SPI continuity to GPIO7, GPIO8, GPIO9, GPIO10, and GPIO11.
- Verify shutdown button toggles GPIO21.
- Verify status LED is connected to GPIO6 through a current-limiting resistor.

11.4 Software Check

- Confirm correct CM4 or CM5 config overlay is installed.
- Confirm serial console is disabled on telemetry UART.
- Confirm I2C devices appear in `i2cdetect`.
- Confirm SPI devices appear as `/dev/spidev0.0` and `/dev/spidev0.1`.
- Confirm MAVLink data is received from Pixhawk.

12 Known Design Considerations

12.1 UART Overlay Dependency

The physical GPIO pins are routed on the PCB, but UART availability depends on the operating system, device tree overlay, and Raspberry Pi platform. The CM4 and CM5 use different internal peripheral mapping. Always verify the final `/dev/ttyAMA*` assignment after boot.

12.2 5 V Back-Power Risk

If the adapter board injects 5 V into the carrier board while USB-C power is also connected, reverse current may occur. Use diode protection, ideal diode OR-ing, or a clearly defined power-source selection jumper.

12.3 I2C Cable Length

I2C is not designed for long cable runs in noisy environments. For drone use, keep I2C wires short and avoid routing them near ESC power lines, motor phase wires, or high-current battery wiring.

12.4 Pixhawk JST GH Pinout

Pixhawk-style TELEM connectors are usually 6-pin JST GH. This adapter board exposes simplified UART connections. A custom harness may be required depending on the flight controller.

13 Recommended Revision Improvements

For the next hardware revision, the following improvements are recommended:

- Add diode or ideal-diode protection on the 5 V input path.
- Add resettable fuse on the external 5 V rail.
- Add ESD protection diodes on external UART, I2C, and SPI connectors.
- Add optional I2C pull-up resistor selection jumpers.
- Add clear connector numbering on both top and bottom silkscreen.
- Add dedicated Pixhawk 6-pin TELEM compatible connector footprint.
- Add LED indicators for 5 V and 3.3 V rail presence.
- Add test pads for all active communication lines.

14 Conclusion

The Universal Drone Adapter Board V4.0 provides a compact and practical companion-computer interface for Raspberry Pi Compute Module based UAV systems. By converting the CM4/CM5 Nano-B 40-pin header into labelled JST GH drone connectors, it simplifies Pixhawk integration, sensor wiring, and payload electronics development.

The board is especially suitable for drone sensor hubs, GPS-denied navigation systems, MAVLink companion computers, optical-flow integration, and modular UAV payload development.

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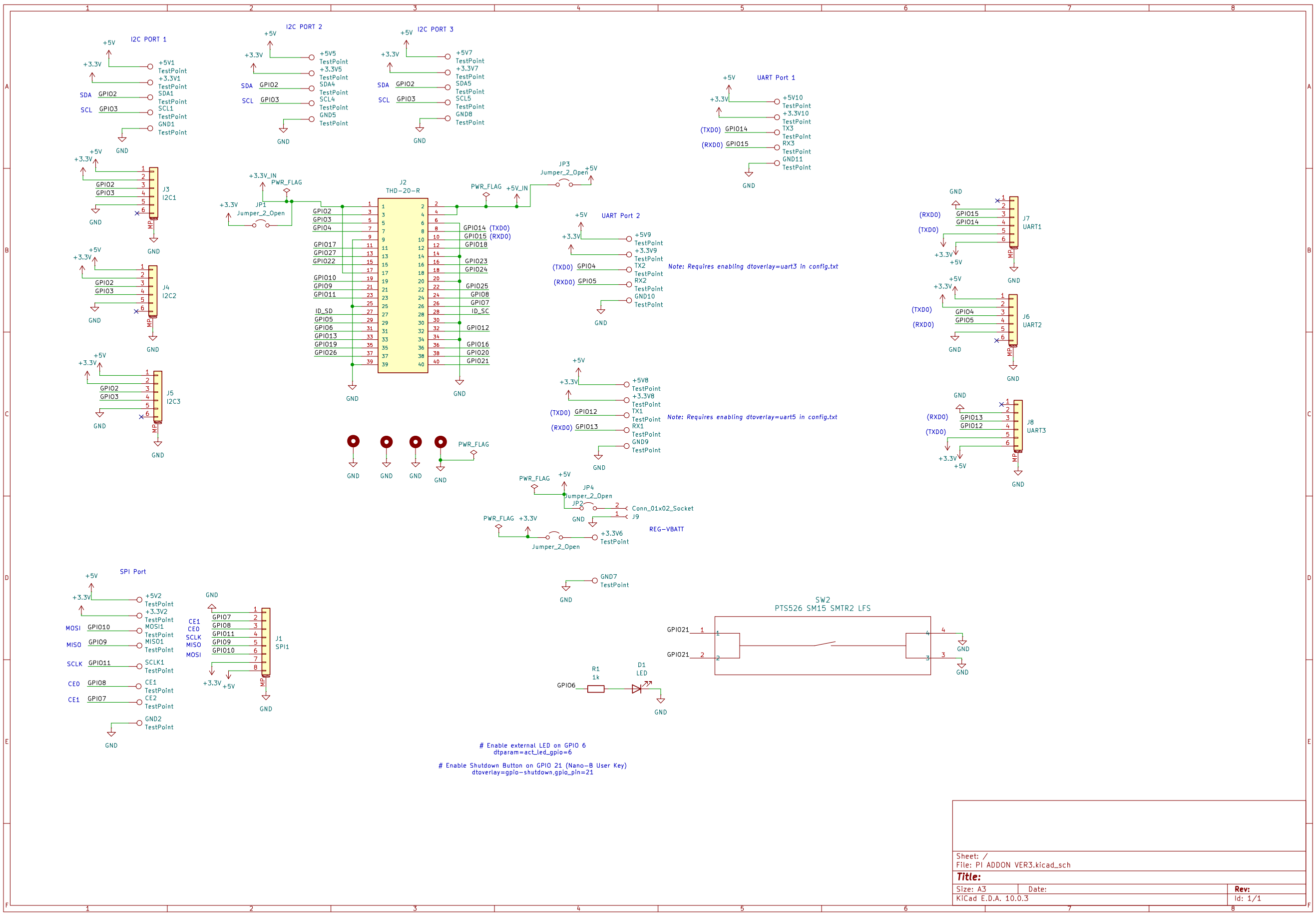

Figure 1: schematic